

Dr. Sonali Sharma

Earth Observation · LiDAR & Point Clouds · GeoAI · Environment & Climate

Goslar, Germany · +49 152 160 44085 · sonali.sharma@web.de

[Portfolio](#) · [LinkedIn](#)



PROFILE

Applied geospatial and remote-sensing scientist (PhD, Environmental Sciences) with 8+ years across industry and research in Finland, Germany and India. I work where Earth observation, LiDAR/point clouds and GeoAI meet real environmental and planning decisions – from automated point-cloud classification and urban tree and biomass inventories to land-cover change, ecosystem services and climate-risk mapping. Looking for a long-term role that combines hands-on geospatial development with technical project ownership and stakeholder-facing delivery.

CORE SKILLS

Earth Observation: Sentinel, Landsat, ASTER DEM, Google Earth Engine; land-cover classification and change detection; land-surface temperature and heat-island mapping; multi-temporal and scenario analysis

LiDAR & 3D: Terrasolid suite and TerraScan macro automation; airborne, mobile and terrestrial point clouds; classification, QC and artefact detection; DTM/DSM/CHM; individual-tree detection and crown delineation; building heights; above-ground biomass and carbon

GeoAI & ML: Python deep learning (CNN, U-Net, transfer learning; reproducible ML pipelines; scikit-learn

GIS & Engineering: R, Python, ArcGIS Pro, QGIS, GeoServer, WFS and spatial databases, HPC, UNIX/Linux, Git/GitHub, reproducible notebooks

Delivery: Interdisciplinary project coordination, stakeholder workshops with municipal and national authorities, technical documentation and reporting

PROFESSIONAL EXPERIENCE

R&D Engineer (LiDAR) | **Terrasolid Oy**, Espoo, Finland

Jul 2026 – present

- Develop and test LiDAR processing workflows for airborne, mobile and terrestrial point-cloud applications across the Terrasolid software suite.
- Design and automate TerraScan macros for point-cloud classification, quality control and detection of artefacts such as low points, multipath errors and glass-roof reflections.
- Build workflows for urban tree inventories and above-ground biomass and carbon estimation, translating research methods into practical, reproducible geospatial solutions.
- Bridge R&D and application: test new methods against real customer data and document them so they can be run repeatably by others.

Postdoctoral Researcher | **University of Helsinki**, Helsinki, Finland

Sep 2023 – Oct 2025

- Designed and automated GIS workflows (Python, R, HPC) integrating airborne LiDAR with high-resolution land-cover data, reducing processing time by approximately 70%.
- Built reproducible machine-learning pipelines for 3D built-and-green urban structure analysis; applied CNN transfer learning to characterise residential typologies at planning scale across the Helsinki Metropolitan Area.
- Derived five LiDAR-based built–green typologies for Nordic residential areas, turning vegetation-structure data into a direct planning tool for identifying which housing types need which vegetation layer.
- Analysed bird-biodiversity data against urban structural heterogeneity to produce planning-relevant spatial evidence on habitat quality.
- Coordinated interdisciplinary delivery with the Finnish Environment Institute (SYKE); presented map products and findings at stakeholder workshops with municipal authorities and at international conferences.

Visiting Doctoral Researcher | **Martin-Luther-Universität Halle-Wittenberg**, Germany *Aug 2020 – Sep 2021*

- Led research on plausible future urban-growth and IPCC climate-change scenarios and their implications for ecosystem services, embedded in German research teams.
- Worked across ArcGIS, Python, R, Google Earth Engine and UNIX/Linux environments; presented at international conferences on sustainability and geospatial methods.

Doctoral Researcher (PhD) | **Jawaharlal Nehru University**, New Delhi, India *Aug 2018 – Jun 2023*

- Modelled urban ecosystem services – soil erosion, flood regulation, heat regulation and carbon sequestration – across two Western Himalayan landscapes, and projected them under combined planning and climate scenarios.
- Integrated multi-source datasets (Landsat/Sentinel, ASTER DEM, climate, socio-economic) to analyse two decades of land-cover change, urban growth dynamics and heat-island effects.
- Translated biophysical models into a spatially explicit flood-risk framework (hazard × exposure × vulnerability) using SCS-CN runoff, Getis-Ord hotspot analysis and WorldPop.
- Mentored junior researchers in GIS workflows and spatial data processing; published four first-author journal articles from the work.

EDUCATION

PhD, Environmental Sciences – Jawaharlal Nehru University, New Delhi	<i>2018 – 2023</i>
MSc, Environmental Sciences – Central University of Rajasthan	<i>2015 – 2017</i>
BSc, Biology – Mohan Lal Sukhadia University	<i>2012 – 2015</i>

LANGUAGES

English (C2, professional working language) · German (A2, B1 course in progress) · Hindi (Native)

SELECTED CONTINUING EDUCATION

- Fundamentals of GeoAI: Deep Learning for Geospatial Analysis – U-Net training for urban image segmentation (2026)
- Urban Analytics in Python (2026) · Spatial Machine Learning and Statistics in Python (2025)
- Location & AI – Basic and Deep Dive into Practical Solutions, Location Innovation Hub (2025)
- Esri MOOCs: Going Places with Spatial Analysis; Make an Impact with Modern Geo Apps (2024–2025)

PUBLICATIONS

6 peer-reviewed · 2 under review · 2 in preparation. Selected:

- Sharma, S., Maija, T., and Andersson, E. (n.d.). Situating private urban green spaces – using transfer learning to characterise urban landscapes. Urban Forestry and Urban Greening.(Revision sent)
- Dennis, M., Huck, J., Holt, C., McHenry, E., Andersson, E., Sharma, S., & Haase, D. (2026). Beyond the patch: leveraging the notion of realized habitat in fragmentation–biodiversity research. Landscape Ecology, 41, 37.
- Sharma, S., Joshi, P.K., & Fürst, C. (2022). Exploring multiscale influence of urban growth on landscape patterns of two emerging urban centres in the Western Himalaya. Land, 11, 2281.
- Sharma, S., Joshi, P.K., & Fürst, C. (2022). Unravelling net primary productivity dynamics under urbanization and climate change in the Western Himalaya. Ecological Indicators, 144, 109508.
- Sharma, S., Sharma, M., Anees, M.M., & Joshi, P.K. (2020). Longitudinal study of changes in ecosystem services in a city of lakes, Bhopal, India. Energy, Ecology and Environment, 6, 408–424.
- Sharma, S., et al. (2020). A long-term and comprehensive assessment of urbanization-induced impacts on ecosystem services in the capital city of India. City and Environment Interactions, 7, 100047.